# AYUSH GOEL


Data Scientist | AI/ML Engineer | AI Automation Engineer

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## PROFESSIONAL SUMMARY


Data Scientist and AI/ML Engineer with hands-on experience deploying intelligent data systems, automation pipelines, and generative AI applications in enterprise environments at Indian Oil Corporation Limited (IOCL), a Fortune 500 organizations. Proficient in the complete data science lifecycle: data collection, EDA, feature engineering, model development, and visualization. Experienced in generative AI (Gemini API), computer vision (OpenCV, MediaPipe), OCR-based extraction, and event-driven automation. NASA Space Apps Challenge 2024 Global Finalist and national ML hackathon top-6 finisher.


## TECHNICAL SKILLS

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|---|---|
| **Machine Learning:** | Supervised/unsupervised learning, regression, classification, feature selection |
| **Deep Learning & AI:** | Neural networks, CNNs, generative AI (Gemini API), LLM integration, NLP basics, prompt engineering, transfer learning |
| **Data Science Stack:** | NumPy, Pandas, Matplotlib, Seaborn, SciPy, scikit-learn, Theano |
| **Computer Vision:** | OpenCV, Tesseract OCR (multi-PSM), MediaPipe Pose Estimation, image preprocessing, face_recognition |
| **Data Visualization:** | Matplotlib, Seaborn, Power BI, Chart.js |
| **Automation & ETL:** | Cron scheduling (node-cron), Nodemailer, event-driven pipelines, ETL, workflow automation |
| **Languages:** | Python (primary), SQL, JavaScript (ES6+), C, C++ |
| **Databases:** | MySQL, SQLite (SQLAlchemy), MongoDB — query optimization, schema design, compound indexing |
| **Platforms & Tools:** | Google Colab, Streamlit, Jupyter Notebook, Power BI, Flask, Django, Git, Postman, VS Code |

## WORK EXPERIENCE

### Information Systems Intern — Data & AI | Indian Oil Corporation Limited (IOCL)

*Aug 2025 – Present*

*Fortune 500 PSU — New Delhi, India*

**OCR Data Extraction Pipeline — Python • OpenCV • Tesseract • Flask • SQLite • ReportLab**

- Developed an end-to-end data extraction pipeline ingesting AC unit images and automatically parsing temperature and humidity sensor readings using multi-strategy OCR, converting unstructured image data into structured, queryable database records.
- Applied 5 image preprocessing techniques (adaptive thresholding, Otsu binarization, histogram equalization, morphological noise reduction, contrast inversion) and evaluated 4 Tesseract PSM configurations (psm 6, 7, 11, 13) per image to maximize extraction accuracy across varied lighting and background conditions.
- Engineered an intelligent metadata extraction layer: EXIF-based timestamp parsing with regex fallback to OCR-based timestamp recovery from cropped image regions (5 crop ratios: 8–20%), with smart error-correction logic for common OCR misreads (e.g., O→0, I→1), producing reliable timestamps for time-series analysis.
- Built an analytical data store in SQLite with Flask REST APIs delivering a paginated history view (100-record limit) with room-based filtering, and automated PDF reporting (ReportLab) with embedded image evidence for compliance documentation.
- Reduced data collection effort from 15+ minutes/day of manual logging to under 10 seconds per reading — achieving 100% automation of routine sensor monitoring with zero human intervention.

**Hardware Analytics Dashboard — Python • Flask • SQLAlchemy • Chart.js**

- Designed a 5-level interactive BI dashboard enabling management to drill from macro (location-level) to micro (individual model) views of 500+ hardware assets across all IOCL sites.
- Built 12+ aggregation API endpoints computing location-wise, type-wise, assignment-status, and employee-wise asset metrics in real time from a normalized 4-table relational schema with strategic indexing.
- Implemented bulk upload and bulk assignment automation via Excel parsing (openpyxl), cutting asset onboarding time by ~80% and delivering actionable insights on assigned vs. unassigned assets to inform procurement decisions.

## PROJECTS

**GYM Intelligence System — Multimodal AI Coaching** *Feb 2025*

*Python • Gemini AI API • MediaPipe • OpenCV • Streamlit*

- Integrated Google Gemini multimodal LLM for nutritional analysis from a single food photo: caloric content, full macronutrient breakdown (protein, carbs, fats, fiber), and a health rating score — demonstrating applied generative AI and prompt engineering skills.
- Applied MediaPipe Holistic Pose Estimation to extract 33-keypoint skeletal representations from live video; computed joint angles (elbow, knee, hip, shoulder) using trigonometric vector math for real-time exercise form classification across 4 exercises: bicep curls, squats, lunges, and shoulder press.
- Built a rule-based ML feedback layer with configurable angular thresholds per exercise, bilateral symmetry detection (left/right imbalance), range-of-motion validation, and consecutive error counting — replicating adaptive personal trainer behavior with escalating audio-visual alerts for form errors and positive reinforcement for correct reps.
- Deployed via Streamlit with live webcam capture, real-time skeleton overlay rendering, rep counter, session accuracy percentage tracker, and a nutrition analysis panel — all in a single unified interface.

**Flight Price Prediction — End-to-End ML Pipeline** *Dec 2024*

*Python • Pandas • NumPy • Matplotlib • Seaborn • scikit-learn • Google Colab*

- Executed a complete supervised ML pipeline on a 2,672-row, 10-feature airline dataset: raw data ingestion, missing value imputation, outlier detection (IQR method), categorical feature encoding (label and one-hot), and normalization.
- Performed multivariate EDA using Seaborn correlation heatmaps, pair plots, and distribution plots to identify the top predictors of flight price: airline carrier, number of stops, departure time bucket, and journey duration.
- Trained and evaluated multiple regression algorithms with hyperparameter tuning; visualized feature importance scores to deliver interpretable, business-friendly insights into model decision-making — demonstrating ML explainability skills.

**Service Payment Alert Automation (PMS)** *Jan 2026*

*Node.js • Express.js • MongoDB • Nodemailer • node-cron • React.js*

- Architected an event-driven automated alert system: cron jobs firing at 10 AM and 4 PM IST daily evaluate all unpaid services against their due dates, classify alerts (Reminder, Overdue, or Renewed), and dispatch personalized HTML emails via Nodemailer with deduplication logic to prevent repeated sends.
- Designed a MongoDB data model (5 collections, compound indexes) optimized for payment trend queries and time-series alert generation across users' recurring service portfolios.
- Built a comprehensive reporting module with statistical summaries (total, paid, unpaid, average payment, frequency distribution) exportable to CSV and PDF for financial analysis.
- Implemented an auto-renewal engine that automatically advances billing cycles for subscriptions with auto-renew enabled, requiring zero manual intervention for recurring service management.

**Exam Proctoring Platform — AI Identity Verification & Behavioral Monitoring** *Jan 2025*

*Django • Python • face_recognition • OpenCV • SQLite • JavaScript*

- Built an AI-powered remote proctoring system with face recognition identity verification at login and at randomized intervals during each examination, comparing live webcam frames against enrollment photos to automatically detect impersonation attempts.
- Automated behavioral surveillance: clipboard operations disabled via JavaScript, window-focus events logged with precise timestamps, and audio activity amplitude sampled every 5 minutes — creating a complete, tamper-evident per-session audit trail reviewable after each exam.
- Built a behavioral anomaly logging system generating structured per-session datasets for post-exam integrity analysis, along with an automated Q&A generation pipeline for professors (objective and subjective question types) and a live student monitoring dashboard.

## EDUCATION

**Diploma in Artificial Intelligence & Machine Learning — Aptech Learning** *2024 – 2025*

*Score: 82% | ML, Deep Learning, NLP, Computer Vision, Data Analytics*

**Bachelor of Business Administration (BBA) — Bharati Vidyapeeth University** *2020 – 2023*

*CGPA: 9.08 / 10.0 | Business Analytics, Research Methodology, Quantitative Methods*

## ACHIEVEMENTS & RECOGNITION

- **NASA Space Apps Challenge 2024 — Global Finalist:** Top global team among 57,000+ participants from 160+ countries; project applied AI/ML to NASA open data for real-world space science problem-solving.
- **ML Quest Hackathon — National Rank 6:** Ranked 6th nationally in a competitive ML hackathon, demonstrating applied ML skills under time-constrained conditions against 200+ teams.
- **ICAI Commerce Wizard — National Position Holder:** Secured an all-India ranking in the ICAI's national-level quantitative and analytical commerce competition.
- **BVP 'Avsar' Business Plan Competition — University Winner:** First place for a data-driven business plan combining market analysis, financial modeling, and technology innovation.